```
#Gen Ai ACknowledgement

I have used Gen AI only for the specific purposes outlined in my acknowledgements

I acknowledge the use of Copilot (Microsoft, https://m365.cloud.microsoft/chat) to
compile my initial notes and to proofread my final draft before submission by the deadline.
```

```
#provides the basemaps that can be used underneath spatial outputs
!pip install contextily
#Instalss spatial plotting tools
!pip install geoplot
#Installs features that allow the addition of a north arrow to maps made
!pip install git+https://github.com/pmdscully/geo_northarrow.git
# imports pandas which is used for data manipulation and gives it a shorter alias which is used in further code.
import pandas as pd
#imports numpy for numerical operations and gives it a shorter alias for use in further code
import numpy as np
#Imports geopandas which is used for spatial data and allows the plotting of maps, points and polygons based on
#what the data shows
import geopandas as gpd
# imports matplotlib for plotting graphs and gives it a shorter alias for use in further code. Plt is used to control
#factors such as titles, labels and display functions of graphs
import matplotlib.pyplot as plt
#Used for adding coordinate reference systems and projections
import pyproj
# imports systems used to add basemaps to spatial plots
import contextily as ctx
# imports seaborn which is used to make nicer statistics visualisations and gives it a shorter alias for use
# in further code.
import seaborn as sns
#importsfeatures that allow for spatial plottng
import geoplot as gplt
# imports features that allow for the manipulation of coordinate referance system
import geoplot.crs as gcrs
# imports a function to add a north arrow to the map
from geo_northarrow import add_north_arrow
```

```
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->geo_northarr
```

```
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->geo_northarr
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->geo_northarrow==0.2.
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->geo_northarro
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from pyogrio>=0.7.2->geopandas->geo_nor
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->g
Building wheels for collected packages: geo_northarrow
  Building wheel for geo_northarrow (pyproject.toml) ... done
  Created wheel for geo_northarrow: filename=geo_northarrow-0.2.0a0-py3-none-any.whl size=14940 sha256=9879f0c9c0ca189379f
  Stored in directory: /tmp/pip-ephem-wheel-cache-brrxzsp2/wheels/d7/28/6c/cb006605c19bc4f58db715626f7113a6858d8c002e8c52f
Successfully built geo_northarrow
Installing collected packages: geo_northarrow
Successfully installed geo_northarrow-0.2.0a0
```

```
#This code opens up a manual file uplode when running the notebook. By using this code i was able to manualy select
#the crime, imd, and lsoa boundry datasets and upload them manually
from google.colab import files
uploaded = files.upload()
#to check the files have been imported correctly and the data is now prepared for future use
#the following code is used to check the directory and make sure they are present
import os
os.listdir()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving LSOA_2021_EW_BGC_V5.cpg to LSOA_2021_EW_BGC_V5.cpg
Saving LSOA_2021_EW_BGC_V5.dbf to LSOA_2021_EW_BGC_V5.dbf
Saving LSOA_2021_EW_BGC_V5.prj to LSOA_2021_EW_BGC_V5.prj
Saving LSOA_2021_EW_BGC_V5.shp to LSOA_2021_EW_BGC_V5.shp
Saving LSOA_2021_EW_BGC_V5.shx to LSOA_2021_EW_BGC_V5.shx
['.config',
 'LSOA_2021_EW_BGC_V5.shx',
 'LSOA_2021_EW_BGC_V5.dbf',
 'IMD_BY_LSOA.csv',
 'LSOA_2021_EW_BGC_V5.shp',
 'LSOA_2021_EW_BGC_V5.cpg',
 'LSOA_2021_EW_BGC_V5.prj',
 'April_Crime_Data.csv',
 'sample_data']
```

```
#this loads the LSOA 2021, IMD and Crime datasets
lsoa = gpd.read_file("LSOA_2021_EW_BGC_V5.shp")
Imd = pd.read_csv("IMD_BY_LSOA.csv")
crime = pd.read_csv("April_Crime_Data.csv")
```

```
#This code shows the first 5 rows of both the IMD and Crime datasets
Imd.head()
crime.head()
```

	Crime ID	Month	Reported by	Falls within	Longitude	Latitude	Location	LSOA code	
0	NaN	2024-04	West Yorkshire Police	West Yorkshire Police	-1.472961	53.599861	On or near Railway Walk	E01007439	Ba
1	317bd0abb97f67a9b5e186e9895d13f4fc99d29f2c9f29...	2024-04	West Yorkshire Police	West Yorkshire Police	-1.732999	53.545103	On or near Wood Royd Hill Lane	E01007426	Ba
2	f3dc2030833b0bc107bc49bfc5182a73b95644e5ed7b3c...	2024-04	West Yorkshire Police	West Yorkshire Police	-1.877569	53.943473	On or near Park Crescent	E01010646	Br
3	9b7f1084be073130cca59d4dfc1cc1cb7fd7909f7c78bb...	2024-04	West Yorkshire Poli	West Yorkshire Poli	-1.875348	53.942581	On or near Sycamore	E01010646	Br

```
# shows the coloumn names, data types, and number of non null values in both datasets
Imd.info()
crime.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 33755 entries, 0 to 33754
Data columns (total 56 columns):
 #   Column                                   Non-Null Count  D
---  ------                                   --------------  -
 0   LSOA code (2021)                         33755 non-null  o
 1   LSOA name (2021)                         33755 non-null  o
 2   Local Authority District code (2024)     33755 non-null  o
```

```
 3   Local Authority District name (2024)                                                        33755 non-null  o
 4   Index of Multiple Deprivation (IMD) Score                                                   33755 non-null  f
 5   Index of Multiple Deprivation (IMD) Rank (where 1 is most deprived)                         33755 non-null  i
 6   Index of Multiple Deprivation (IMD) Decile (where 1 is most deprived 10% of LSOAs)          33755 non-null  i
 7   Income Score (rate)                                                                         33755 non-null  f
 8   Income Rank (where 1 is most deprived)                                                      33755 non-null  i
 9   Income Decile (where 1 is most deprived 10% of LSOAs)                                       33755 non-null  i
 10  Employment Score (rate)                                                                     33755 non-null  f
 11  Employment Rank (where 1 is most deprived)                                                  33755 non-null  i
 12  Employment Decile (where 1 is most deprived 10% of LSOAs)                                   33755 non-null  i
 13  Education, Skills and Training Score                                                        33755 non-null  f
 14  Education, Skills and Training Rank (where 1 is most deprived)                              33755 non-null  i
 15  Education, Skills and Training Decile (where 1 is most deprived 10% of LSOAs)               33755 non-null  i
 16  Health Deprivation and Disability Score                                                     33755 non-null  f
 17  Health Deprivation and Disability Rank (where 1 is most deprived)                           33755 non-null  i
 18  Health Deprivation and Disability Decile (where 1 is most deprived 10% of LSOAs)            33755 non-null  i
 19  Crime Score                                                                                 33755 non-null  f
 20  Crime Rank (where 1 is most deprived)                                                       33755 non-null  i
 21  Crime Decile (where 1 is most deprived 10% of LSOAs)                                        33755 non-null  i
 22  Barriers to Housing and Services Score                                                      33755 non-null  f
 23  Barriers to Housing and Services Rank (where 1 is most deprived)                            33755 non-null  i
 24  Barriers to Housing and Services Decile (where 1 is most deprived 10% of LSOAs)             33755 non-null  i
 25  Living Environment Score                                                                    33755 non-null  f
 26  Living Environment Rank (where 1 is most deprived)                                          33755 non-null  i
 27  Living Environment Decile (where 1 is most deprived 10% of LSOAs)                           33755 non-null  i
 28  Income Deprivation Affecting Children Index (IDACI) Score (rate)                            33755 non-null  f
 29  Income Deprivation Affecting Children Index (IDACI) Rank (where 1 is most deprived)         33755 non-null  i
 30  Income Deprivation Affecting Children Index (IDACI) Decile (where 1 is most deprived 10% of LSOAs)  33755 non-null  i
 31  Income Deprivation Affecting Older People (IDAOPI) Score (rate)                             33755 non-null  f
 32  Income Deprivation Affecting Older People (IDAOPI) Rank (where 1 is most deprived)          33755 non-null  i
 33  Income Deprivation Affecting Older People (IDAOPI) Decile (where 1 is most deprived 10% of LSOAs)   33755 non-null  i
 34  Children and Young People Sub-domain Score                                                  33755 non-null  f
 35  Children and Young People Sub-domain Rank (where 1 is most deprived)                        33755 non-null  i
 36  Children and Young People Sub-domain Decile (where 1 is most deprived 10% of LSO            33755 non-null  i
 37  Adult Skills Sub-domain Score                                                               33755 non-null  f
 38  Adult Skills Sub-domain Rank (where 1 is most deprived)                                     33755 non-null  i
 39  Adult Skills Sub-domain Decile (where 1 is most deprived 10% of LSOAs)                      33755 non-null  i
 40  Geographical Barriers Sub-domain Score                                                      33755 non-null  f
 41  Geographical Barriers Sub-domain Rank (where 1 is most deprived)                            33755 non-null  i
 42  Geographical Barriers Sub-domain Decile (where 1 is most deprived 10% of LSOAs)             33755 non-null  i
 43  Wider Barriers Sub-domain Score                                                             33755 non-null  f
 44  Wider Barriers Sub-domain Rank (where 1 is most deprived)                                   33755 non-null  i
 45  Wider Barriers Sub-domain Decile (where 1 is most deprived 10% of LSOAs)                    33755 non-null  i
 46  Indoors Sub-domain Score                                                                    33755 non-null  f
 47  Indoors Sub-domain Rank (where 1 is most deprived)                                          33755 non-null  i
 48  Indoors Sub-domain Decile (where 1 is most deprived 10% of LSOAs)                           33755 non-null  i
 49  Outdoors Sub-domain Score                                                                   33755 non-null  f
 50  Outdoors Sub-domain Rank (where 1 is most deprived)                                         33755 non-null  i
 51  Outdoors Sub-domain Decile (where 1 is most deprived 10% of LSOAs)                          33755 non-null  i
```

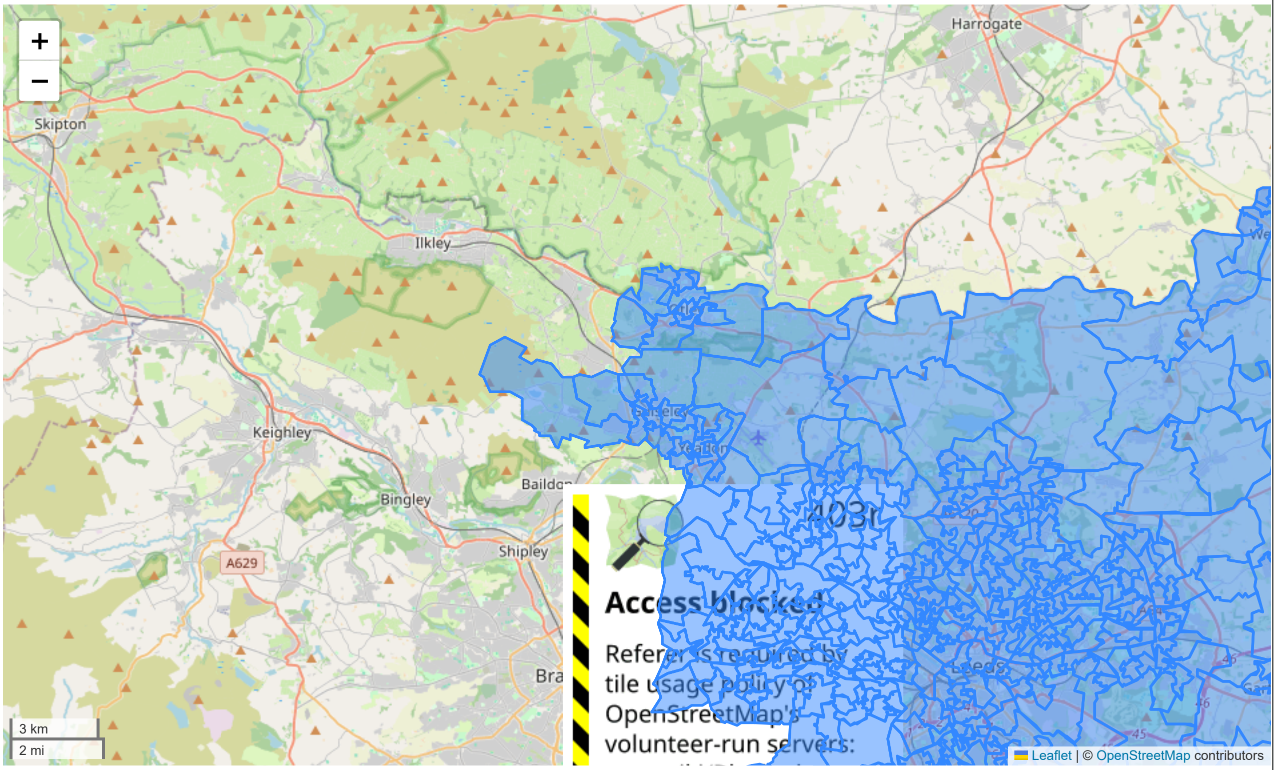
```
#shows the names of the colomns within the IMD dataset
Imd.columns
```

```
Index(['LSOA code (2021)', 'LSOA name (2021)',
       'Local Authority District code (2024)',
       'Local Authority District name (2024)',
       'Index of Multiple Deprivation (IMD) Score',
       'Index of Multiple Deprivation (IMD) Rank (where 1 is most deprived)',
       'Index of Multiple Deprivation (IMD) Decile (where 1 is most deprived 10% of LSOAs)',
       'Income Score (rate)', 'Income Rank (where 1 is most deprived)',
       'Income Decile (where 1 is most deprived 10% of LSOAs)',
       'Employment Score (rate)', 'Employment Rank (where 1 is most deprived)',
       'Employment Decile (where 1 is most deprived 10% of LSOAs)',
       'Education, Skills and Training Score',
       'Education, Skills and Training Rank (where 1 is most deprived)',
       'Education, Skills and Training Decile (where 1 is most deprived 10% of LSOAs)',
       'Health Deprivation and Disability Score',
       'Health Deprivation and Disability Rank (where 1 is most deprived)',
       'Health Deprivation and Disability Decile (where 1 is most deprived 10% of LSOAs)',
       'Crime Score', 'Crime Rank (where 1 is most deprived)',
       'Crime Decile (where 1 is most deprived 10% of LSOAs)',
       'Barriers to Housing and Services Score',
       'Barriers to Housing and Services Rank (where 1 is most deprived)',
       'Barriers to Housing and Services Decile (where 1 is most deprived 10% of LSOAs)',
       'Living Environment Score',
       'Living Environment Rank (where 1 is most deprived)',
       'Living Environment Decile (where 1 is most deprived 10% of LSOAs)',
       'Income Deprivation Affecting Children Index (IDACI) Score (rate)',
       'Income Deprivation Affecting Children Index (IDACI) Rank (where 1 is most deprived)',
       'Income Deprivation Affecting Children Index (IDACI) Decile (where 1 is most deprived 10% of LSOAs)',
       'Income Deprivation Affecting Older People (IDAOPI) Score (rate)',
       'Income Deprivation Affecting Older People (IDAOPI) Rank (where 1 is most deprived)',
       'Income Deprivation Affecting Older People (IDAOPI) Decile (where 1 is most deprived 10% of LSOAs)',
       'Children and Young People Sub-domain Score',
       'Children and Young People Sub-domain Rank (where 1 is most deprived)',
       'Children and Young People Sub-domain Decile (where 1 is most deprived 10% of LSO',
       'Adult Skills Sub-domain Score',
       'Adult Skills Sub-domain Rank (where 1 is most deprived)',
       'Adult Skills Sub-domain Decile (where 1 is most deprived 10% of LSOAs)',
```

```
         'Geographical Barriers Sub-domain Score',
         'Geographical Barriers Sub-domain Rank (where 1 is most deprived)',
         'Geographical Barriers Sub-domain Decile (where 1 is most deprived 10% of LSOAs)',
         'Wider Barriers Sub-domain Score',
         'Wider Barriers Sub-domain Rank (where 1 is most deprived)',
         'Wider Barriers Sub-domain Decile (where 1 is most deprived 10% of LSOAs)',
         'Indoors Sub-domain Score',
         'Indoors Sub-domain Rank (where 1 is most deprived)',
         'Indoors Sub-domain Decile (where 1 is most deprived 10% of LSOAs)',
         'Outdoors Sub-domain Score',
         'Outdoors Sub-domain Rank (where 1 is most deprived)',
         'Outdoors Sub-domain Decile (where 1 is most deprived 10% of LSOAs)',
         'Total population: mid 2022', 'Dependent Children aged 0-15: mid 2022',
         'Older population aged 60 and over: mid 2022',
         'Working age population 18-66 (for use with Employment Deprivation Domain): mid 2022'],
        dtype='object')
```

```
# # Data downloaded from https://geoportal.statistics.gov.uk/maps/761ecd09b4124843b95511a242e2b1a1
#this code loads the 2021 LSOA boundry shapefile using geopandas and the dataset contains the actual polygon
#data for all LSOAs in England
shp =gpd.read_file('LSOA_2021_EW_BGC_V5.shp')
#to create a subset of just the study area a filter was applied to extract only the LSOAs that had a name containing
#the city 'leeds'
leeds_shp =shp.loc[shp['LSOA21NM'].str.contains('Leeds'),:]
#the filtered dataset is then saved for use in later coding
leeds_shp.to_file('Leeds.geojson')
```

```
#this code provides an interactive map preview of the Leeds LSOAs
leeds_shp.explore()
```




```
#shows the metadata for the file including number of LSOAs with leeds in their name, the amount of rows and coloums and ob
#for each column
leeds_shp.info()
```

```
<class 'geopandas.geodataframe.GeoDataFrame'>
Index: 488 entries, 10719 to 33048
Data columns (total 9 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   LSOA21CD   488 non-null    object
 1   LSOA21NM   488 non-null    object
 2   LSOA21NMW  0 non-null      object
 3   BNG_E      488 non-null    int64
 4   BNG_N      488 non-null    int64
 5   LAT        488 non-null    float64
 6   LONG       488 non-null    float64
 7   GlobalID   488 non-null    object
 8   geometry   488 non-null    geometry
dtypes: float64(2), geometry(1), int64(2), object(4)
memory usage: 38.1+ KB
```

```
#checks the coordinate system of the boundry dataset so that we know what crs to change the other files to
#as they need to match to have correct analysis later
leeds_shp.crs
```

```
<Projected CRS: EPSG:27700>
Name: OSGB36 / British National Grid
Axis Info [cartesian]:
- E[east]: Easting (metre)
- N[north]: Northing (metre)
Area of Use:
- name: United Kingdom (UK) - offshore to boundary of UKCS within 49°45'N to 61°N and 9°W to 2°E; onshore Great Britain
(England, Wales and Scotland). Isle of Man onshore.
- bounds: (-9.01, 49.75, 2.01, 61.01)
Coordinate Operation:
- name: British National Grid
- method: Transverse Mercator
Datum: Ordnance Survey of Great Britain 1936
- Ellipsoid: Airy 1830
- Prime Meridian: Greenwich
```

```
#This block converts the crime dataset from a standard dataframe to a geodataframe and creates
#point geomitries from the longitude and latitude column. This allows the crime data to be spatially aggrigated
#and plotted on the map of leeds later on. It was given a crs of 4326 as this is the universal standard
#when converting gps coordinates
crime_gdf = gpd.GeoDataFrame(
    crime, geometry=gpd.points_from_xy(crime.Longitude, crime.Latitude))
crime_gdf = crime_gdf.set_crs(epsg = 4326)
#the crs of crime data was then reprojected to match that of the Leeds Lsoa which means any spatial joins
#will now be accurate
crime_gdf = crime_gdf.to_crs(leeds_shp.crs)
```

```
#This code does a spatial join containing the point data juat made and the polygon boundries of LSOAs
#the 'within' precidate ensures that only the crimes located inside of an LSOA polygon are matched up
crime_lsoa = gpd.sjoin(crime_gdf, leeds_shp, how="inner", predicate="within")
```

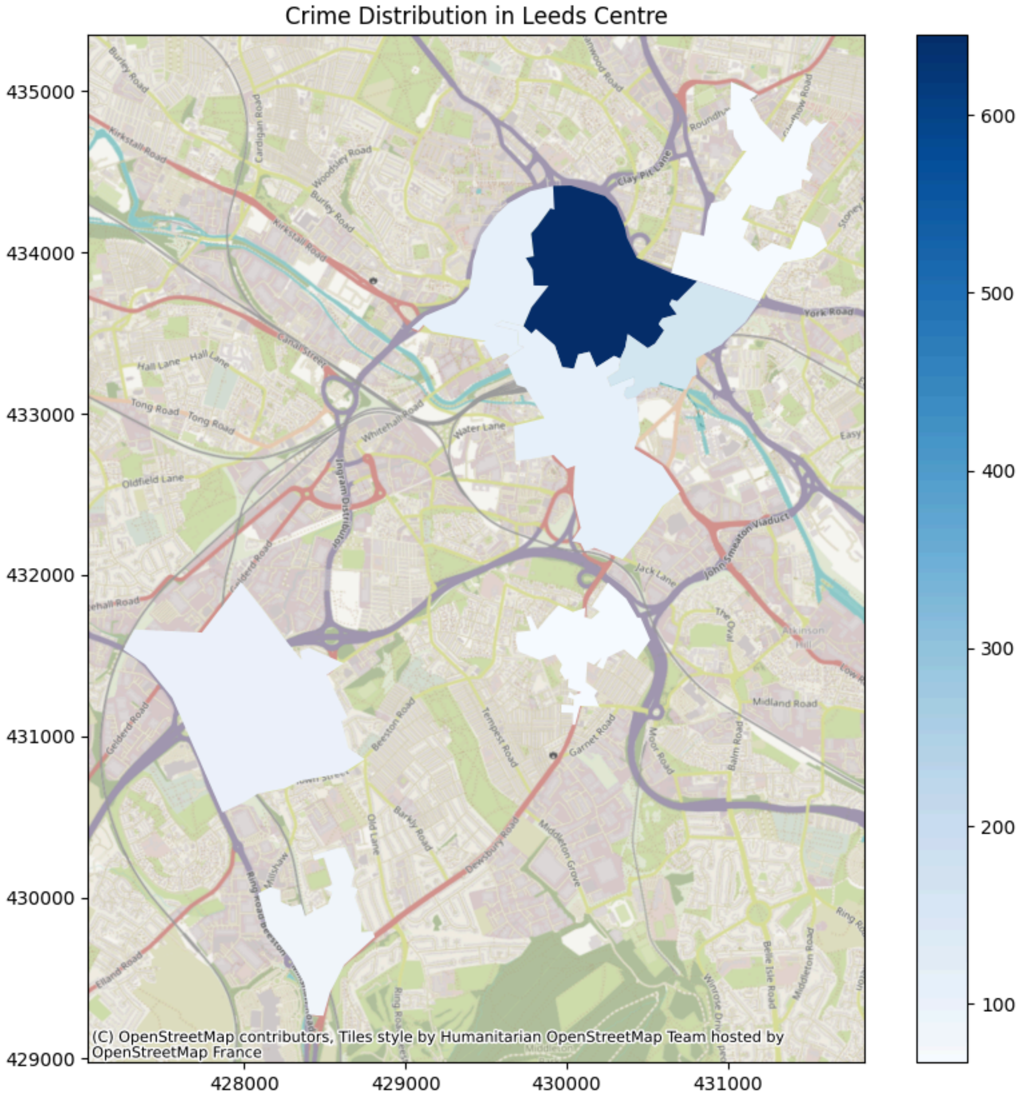
```
#This block groups the joined datasets by the LSOA code using 'LSOA21CD' with size counting the number of incidents in each
#whilst then resenting the result into a clean dataframe with a new column called 'crime count' being added.
#the counts are then merged back into the boundries dataframe so that each LSOA has their associated total
#number of crimes. Finally 'fillna(0)'was used to ensure that LSOAs with no crime records are given a value of 0 instead of
#data all together
crime_counts = crime_lsoa.groupby("LSOA21CD").size().reset_index(name="crime_count")
leeds_shp = leeds_shp.merge(crime_counts, on="LSOA21CD", how= "left")
leeds_shp["crime_count"] = leeds_shp["crime_count"].fillna(0)
```

```
#this code merges the IMD dataset to the LSOA dataset which in turn adds deprivation scores to to each LSOA enabling compar
#and socioeconomic conditions
leeds_shp = leeds_shp.merge(Imd, left_on="LSOA21CD", right_on="LSOA code (2021)", how="left")
```

```
#This code filters the dataset to identify LSOAs with more than 65 crimes
centre_lsoa=leeds_shp.loc[leeds_shp["crime_count"]> 65,:]
```

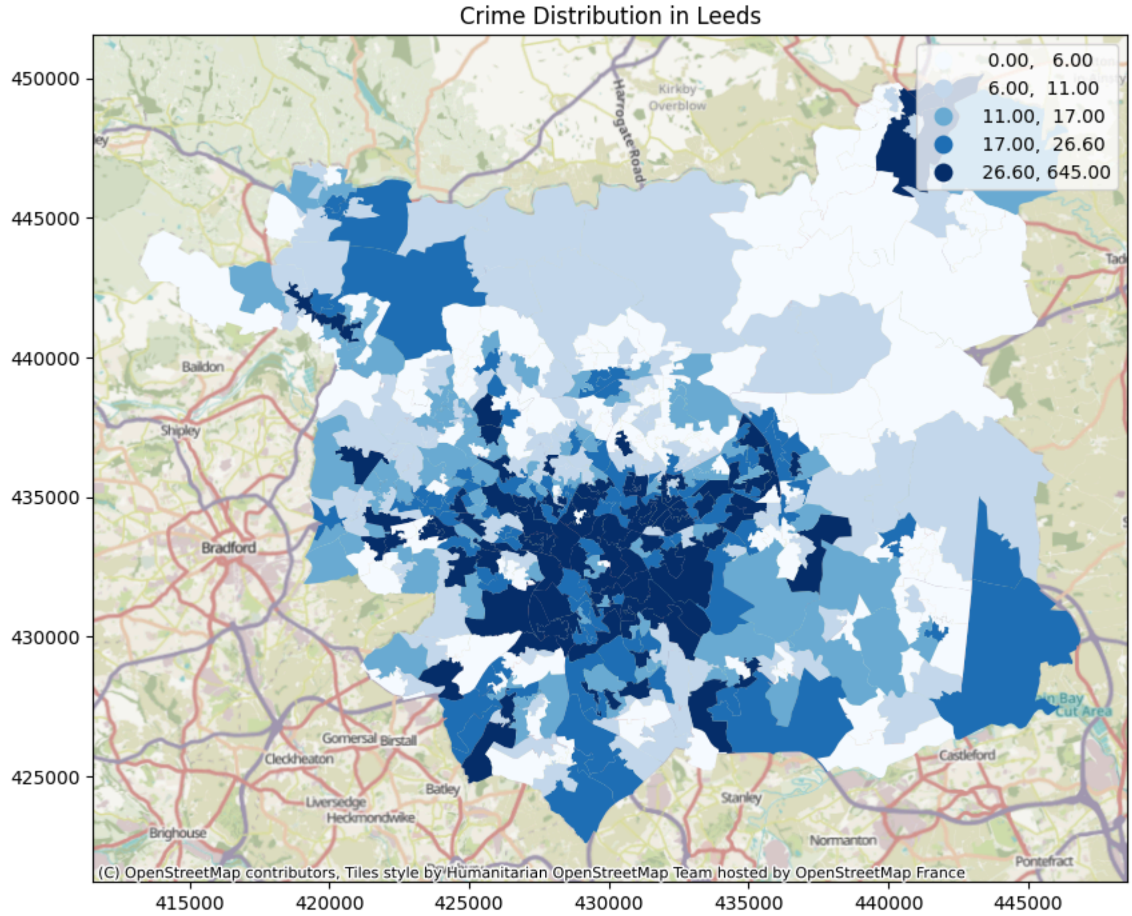
```
#This block produces a choropleth map with the polygons coloured by crime level, with blue beimg used for the colour
#as it is colourblind friendly and is a sequential colourscheme. The map also has a legend to display the
#colour scale and allows for easy interpritation. A basemap was then added using the same crs as the LSOAs to ensure spati
#uniformity and context. The map overall highlights the spatial concentration of crime in central Leeds.
f,ax = plt.subplots(1,figsize=(10, 10))
centre_lsoa.plot( ax=ax, column= "crime_count", cmap= "Blues",legend= True )
ctx.add_basemap(ax,crs= leeds_shp.crs)
plt.title("Crime Distribution in Leeds Centre")
```

```
Text(0.5, 1.0, 'Crime Distribution in Leeds Centre')
```



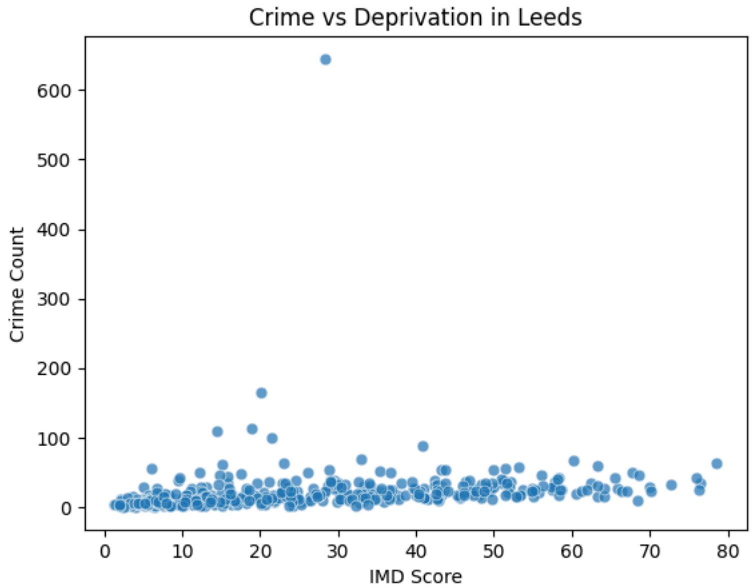

```
#This block of code visualises crime counts across all the Leeds LSAOs with it being arranged in a quantiles fashion which
#devides the data into five equal sized groups ensuring fair colour distribution.This method is usefull when a dataset
#can be unevenly distributed. The title and basemap here improve readability and comunication for all readers.
f,ax = plt.subplots(1,figsize=(10, 10))
leeds_shp.plot( ax=ax, column= "crime_count", cmap= "Blues",scheme="quantiles",k=5,legend= True )
ctx.add_basemap(ax,crs= leeds_shp.crs)
plt.title("Crime Distribution in Leeds")
```

```
Text(0.5, 1.0, 'Crime Distribution in Leeds')
```




```
#This scatterplot examines the relationship between crime counts and IMD score for the LSOAs. The use of the 'virdis' palle
#uniform and clear despite the slight transparency that the individual points have. This visualisation helps identifywhethe
#experience higher crime levels.
sns.scatterplot(data= leeds_shp, x="Index of Multiple Deprivation (IMD) Score", y= "crime_count", palette='viridis', marker
plt.title("Crime vs Deprivation in Leeds")
plt.xlabel("IMD Score")
plt.ylabel("Crime Count")
```

```
/tmp/ipykernel_31752/2462342411.py:4: UserWarning: Ignoring `palette` because no `hue` variable has been assigned.
  sns.scatterplot(data= leeds_shp, x="Index of Multiple Deprivation (IMD) Score", y= "crime_count", palette='viridis', marke
Text(0, 0.5, 'Crime Count')
```




```
Analysis Introduction:
This study investigates the spatial distribution of crime across Leeds and
its relationship with socioeconomic deprivation as understanding where crime is concentrated, alongside the factors associa
essential for informing effective police strategies and improving public safety. In particular, deprivation has been linkec
an important variable for spatial analysis. By using open-source data, including police record crime incidents and the inde
spatial data science techniques on python to analyse crime and deprivation patterns by Lower Layer Super Output Area (LSOA)
```

deprivation measures, the analysis aims to identify if more deprived areas experience higher levels of crime. The findings
policy makers and local authorities in targeting police resources more effectively, contributing to safer and more equitab

Analysis Data:
This study utilises three open-source datasets to analyse the relationship between crime and deprivation in Leeds whereby c
and consists of recorded crime incidents for April 2024. The dataset included geographic coordinates (Longitude and Latitu
incident to be spatially represented as a point. Socioeconomic conditions were measured using the IMD dataset for 2025, wh
in England. The IMD incorporates multiple dimensions including income, health, employment, education, and housing making i
Spatial boundaries were defined using the 2021 LSOA boundary dataset for England and Wales meaning the polygons represente
enable reliable areas to be given to the point-based crime data.
Before carrying out the main analysis an exploratory data analysis was conducted to understand the structure, completeness
being explored with .head and .info to confirm key variables such as coordinates alongside LSOA codes and make sure they w
The LSOA dataset was used alongside .explore to produce a visually interactive map which confirmed the range of the study
revealed that there was early evidence of spatial clustering around the centre of Leeds and that some LSOAs reported zero

Analysis Methods:
Initial preprocessing involved importing the required datasets and inspecting their structure using summary functions, lik
potential data quality issues. The crime dataset was examined to ensure that latitude and longitude fields were formatted c
were removed to retain analytical reliability. Alongside this, the IMD dataset was explored to identify the columns used w
that were required later for merging. In order to define the study area the LSOA boundry dataset was examined and filter
a geospatial dataset representing the study area which was used in further analysis.
With crime data being provided as point coordinates, they were converted into a Geodataframe to enable spatial analysis wi
as EPSG:4326 which is the standard for longitude and latitude data (World Geodetic System, 1984) and was then reprojected
seamless spatial alterations. Following this a spatial join was performed to assign each crime event to an LSOA based on g
it transforms point based data into area based data, allowing for comparison of crime rates between LSOAs which allows fo
highest crime rate. The use of the predicate 'within' ensures that only crimes located within the LSOA boundaries were in
Following the spatial join, crime incidents were managed using a group by function that calculated the total number of cri
'crime count', representing the frequency of crimes per LSOA. The grouped counts were then merged back into the LSOA Geod
those with zero record crimes alongside any missing values being replaced with 0 through .fillna (0) which ensured data c
the spatial data through the same means as the crime events which in turn enabled the interrogation of deprivation scores
socioeconomic conditions and crime levels.
To explore spatial patterns, a choropleth map was produced using the crime count data with the map styled in a sequential c
people are able to visualise the variations in crime intensity across Leeds. Additionally a scatterplot was created to exa
scores which is a reliable and efficient way to demonstrate any trends or correlations within the datasets. Together thes
spatial relationships and generate useful insights into crime policy planning and urban development plans.

Analysis Results:
The spatial distribution of crime across Leeds reveals a clear pattern of concentration within central urban areas with th
in and around Leeds city centre display the highest crime counts, while peripheral areas display significantly lower amou
occurred with intensity decreasing outward from the centre which is visually displayed through the darker shading of cent
rural areas. The scatterplot examining the relationship between crime count and deprivation score suggests a positive ass
scores generally tend to show higher crime counts, although the relationship is not perfectly linear. There is noticeable
moderate levels of crime, and some highly deprived areas showing lower than expected crime counts. Overall the results in
play a role in shaping crime patterns across Leeds, with centrality and deprivation emerging as key influential factors.
The choropleth map created through the analysis is intended for policymakers and local authorities who dictate where polic
overview of crime distribution in Leeds so they can easily see where resources should be prioritised. The scatterplot is i
statistical relationships over geographic outputs which allows for an analysis into whether deprivation scores impact cri

Analysis Discussion:
The results indicate a clear spatial connection between crime within central Leeds, alongside a general positive relations
counts in the city centre is likely influenced by increased population density, economic activity and accessibility (Chat
of businesses, retail areas and transport hubs, all of which attract larger numbers of people and consequently provide m
established urban theories where crime rates are associated with areas of high activity and movement. The observed relat
research which suggests that socioeconomic disadvantage can contribute to higher crime rates as areas with higher IMD s
crime counts (Sampson, et al. 1997). This may reflect underlying factors such as unemployment, reduced access to resour
both committing and experiencing crime. However, given the relationship in the scatterplot is not perfectly linear it
for crime patterns.
Some less deprived areas still displayed moderate crime levels particularly those located near the city centre which in tu
that proximity to high activity areas can heavily influence socioeconomic conditions. Conversely some highly deprived area
or differences in community dynamics (Meier, 1966). In terms of analytical limitations to the study the crime data only r
crime trends may not be captured effectively in this analysis. Secondly the IMD dataset displays deprivation at a differ
lead to some inaccurate results within the study. The same could be said for recorded crime data also as it could have
lead to either an overestimate or underestimate in crime results (White and hansel, 1996).

Analysis Conclusion:
This study analysed the spatial distribution of crime in Leeds and its relationship with socioeconomic deprivation using s
crime is strongly concentrated within central urban areas, with significantly lower levels observed in surrounding areas.
deprivation and crime was identified suggesting that more deprived areas had higher crime rates. However, the findings al
factors such as spatial and urban, rather than just deprivation alone which emphasises the importance of both geographic
public policy perspective these insights could support more targeted policing strategies and resource allocation particu
value of integrating spatial and statistical analysis to better understand complex urban issues and inform decision mak